NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Senior Python_FastAPI & SQL_DBA Specialist • R&D Labs #1

+91 6305761091 • saratchandranallabati@gmail.com • Portfolio: sarat-2007.github.io • GitHub: github.com/sarat-2007 • LinkedIn

Education & National Merit Pedigree

Amrita Vishwa Vidyapeetham — **August 2024 – Expected May 2028**

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort) — *Amritapuri, India*

- **Academic Standing: AP POLYCET State Merit Rank: 497** (Top 0.5% Statewide Percentile).
- **National Distinctions: ISRO-IIRS YUVIKA** (Selected for space science analytics) | **NAEST National Physics Test** (Cleared Prelims & Finals, IAPT / Prof. H.C. Verma) | **MTSO Olympiad** (Level 2 Qualifier).

Peer-Reviewed Publications & Proceedings

- **N. V. D. S. Sarat Chandra**, M. Nunna, Asha C. A., P. C. Sreekumar, **"Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity."** in *Proc. IEEE ICRM*, 2025.
- **N. V. D. S. Sarat Chandra**, et al., **"Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation."** Submitted to *IEEE IES*, 2026.
- **N. V. D. S. Sarat Chandra**, et al., **"Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty."** Submitted 2026.

Targeted Engineering Projects & Systems Architecture

Blood Neural Network: PINNs for Inverse Hemodynamic Rheology Discovery — **SciML Researcher**

PINNs, Navier-Stokes, Carreau-Yasuda, ANSYS Fluent CFD, NTK Trace Balancing

- **Core Impact:** <2.4% relative L2 error vs high-resolution ANSYS Fluent pulsatile CFD benchmarks.

Domain-Aware Industrial Failure Prediction & Diagnostics Framework — **PIML Researcher / Data Science Lead**

Pandas, NumPy, Optuna Bayesian Ensemble, Conformal Uncertainty, XGBoost

- **Core Impact:** 85.2% recall at 91.4% precision (0.94 AUROC) across 120k telemetry time-series samples.

Multi-Parameter Health Vitals Monitor & Real-Time IoT Data Streamer — **Biomedical Systems Developer**

MAX30102 PPG, ESP8266/ESP32, Fast-Mode I2C, 2nd-order IIR Bandpass, FreeRTOS

- **Core Impact:** <200ms real-time SpO2/HRV cloud telemetry; 9-clock I2C bus lockup recovery.

Technical Skills & Core Competencies

Core Languages & Math: Python (3.12), SQL, C++, MATLAB, Multivariable Calculus, Linear Algebra, Probability

Data Science & ML Stack: Pandas, NumPy, Scikit-Learn, Optuna (Bayesian Optimization), PyTorch, Conformal Uncertainty

Data Pipelines & Time-Series: EDA, Time-Series Feature Engineering, ETL Ingestion, REST APIs, JSON/Parquet, SVD/POD

Analytics & Tooling: Jupyter, Git/GitHub, Docker, Linux (Ubuntu Server), ANSYS Fluent, LaTeX, Statistical Testing